

Trainings ▾

General Information

Cummins Inc. recommends use of fully formulated antifreeze or coolant containing a precharge of supplemental coolant additive (SCA). Antifreeze or coolant **must** meet specifications outlined in The Maintenance Council, Recommended Practice 329 (ethylene glycol) or Recommended Practice 330 (propylene glycol). Using fully formulated antifreeze or coolant significantly simplifies cooling system maintenance.

Copies of The Maintenance Council specifications can be obtained by contacting:

- The Maintenance Council
- American Trucking Association
- 2200 Mill Road
- Alexandria, VA 22314-5388
- Phone: (703) 838-1763
- Fax: (703) 836-6070

Fully formulated antifreeze contains balanced amounts of antifreeze, SCA, and buffering compounds, but does **not** contain 50 percent water. Fully formulated coolant contains balanced amounts of antifreeze, SCA, and buffering compounds already premixed 50/50 with deionized water.

The following pages will give an explanation of water, antifreeze, and SCAs. They will also explain how to test antifreeze and SCA levels.

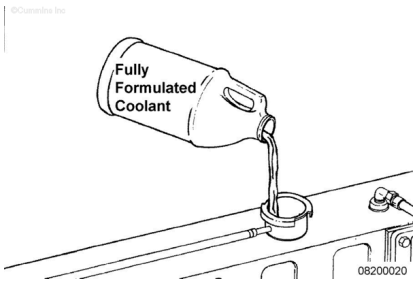
This section also contains information on cooling system maintenance and a coolant treatment chart that is used to determine the correct SCA service filter.

Alternative maintenance practices for cooling systems can be found in Fluids for Cummins® Products Service Manual, Bulletin 5411406.

Fully Formulated Coolant/Antifreeze

Cummins Inc. recommends using either a 50/50 mixture of good-quality water and fully formulated antifreeze, or fully formulated coolant when filling cooling system. The fully

formulated antifreeze or coolant **must** meet The Maintenance Council, Recommended Practice 329 or Recommended Practice 330 specifications.



Good-quality water is important for cooling system performance. Excessive levels of calcium and magnesium contribute to scaling problems, and excessive levels of chlorides and sulfates cause cooling system corrosion.

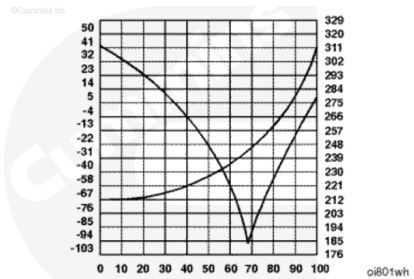
Water Quality	
Calcium Magnesium (Hardness)	Maximum 170 ppm as (CaCO ₃ + MgCO ₃)
Chloride	40 ppm as(Cl)
Sulfur	100 ppm as (SO ₄)

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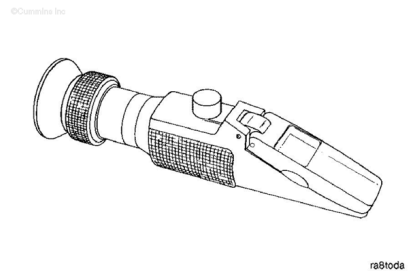
Cummins Inc. recommends use of Fleetguard® Compleat™. It is available in both glycol forms (ethylene and propylene) and complies with The Maintenance Council standards.



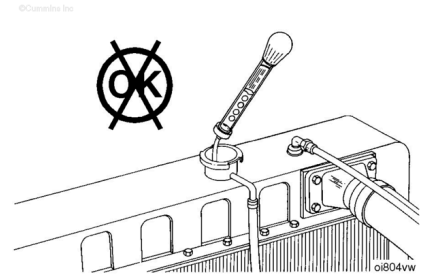
Fully formulated antifreeze **must** be mixed with quality water at a 50/50 ratio (40-percent to 60 percent working range). A 50/50 mixture of antifreeze and water gives a -36°C [-34°F] freezing point and a 110°C [228°F] boiling point, which is adequate for locations in North America. The lowest freezing point of ethylene glycol antifreeze (-173°C [-94°F]) is obtained at 68 percent. Use of higher concentrations of antifreeze will raise freezing point of solution and increase possibility of a silicate gel problem.



Refractometer, Part Number CC-2800, **must** be used to measure freezing point of coolant accurately.



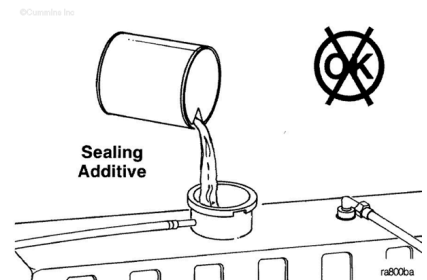
Do **not** use a floating ball hydrometer. Use of a floating ball hydrometer can give an incorrect reading.



Cooling System Sealing Additives

Do **not** use sealing additives in the cooling systems. The use of sealing additives will:

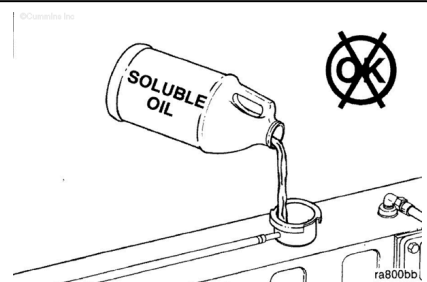
- Build up in coolant low-flow areas
- Clog coolant filters
- Plug radiator and oil coolers
- Plug fuel coolers.



Cooling System Soluble Oils

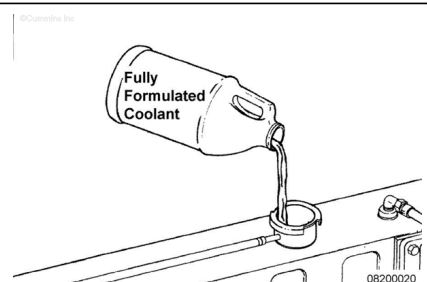
Do **not** use soluble oils in cooling system. Use of soluble oils will:

- Allow cylinder liner pitting
- Corrode brass and copper
- Damage heat transfer surfaces
- Damage seals and hoses.



Supplemental Coolant Additive (SCA)

Fully formulated products contain SCAs and are required to protect cooling system from fouling, solder blooming, and general corrosion. Coolant filter is required to protect the cooling system from abrasive materials, debris, and precipitated coolant additives.



SCAs, or equivalent, are used to prevent liner pitting, corrosion, and scale deposits in cooling system.

Use correct Fleetguard® coolant filter to maintain recommended SCA concentration in system.

Maintain correct concentration by changing service filter at each oil drain interval.

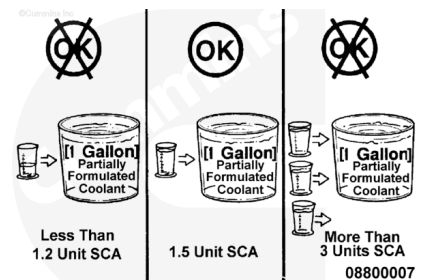
Note : The correct filter is determined by total cooling system capacity and oil drain interval. Reference Coolant Capacity charts. See equipment manufacturer service information for total cooling system capacity. See procedure 018-024 in Section 18 for filter specifications.



⚠ CAUTION ⚠

Insufficient concentration of coolant additives will result in liner pitting and engine malfunction.

The SCA concentration **must not** fall below 1.2 units or exceed 3 units per 3.8 liters [1 gal] of cooling system capacity.



Testing SCA Concentration Level CC-2602 Test Kit

Carefully follow these instructions to test coolant and take appropriate action recommended by kit.

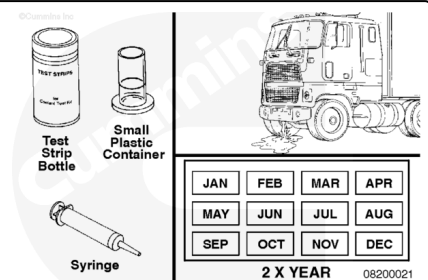
- The coolant sample to be tested **must** be between 10 and 54°C [50 and 130°F]. If sample is too cold or too hot, you will get incorrect results.
- To get best color match results, compare test strip pads to color chart in daylight or under cool white fluorescent lighting. If unsure about a specific color match when a test does fall between two colors on color chart, choose lower numbered block. It is safer to underestimate your results than to overestimate.
- The test strips have a limited shelf life and are sensitive to humidity and extreme heat. Proper handling and storage is necessary to protect life of strips.
- Keep cap tightly sealed on test strip bottle except when removing a strip. Store away from direct sunlight and in an area where temperature will generally stay below 32°C [90°F].
- Do **not** use test strips after the expiration date stamped on bottle.
- Discard kit if any of the pads on unused strips have turned light brown or pink.

- Use one strip at a time and take care **not** to touch any of pads on the strip. Doing so will contaminate the pads and affect test results.
- If strip container is left uncapped for 24 hours, moisture in air will render strips useless, even though **no** discoloration will be evident.
- **Only** use color chart supplied with kit.
- Clean and dry sample cup and syringe after each use. This will prevent contaminating future samples.
- Following correct test times is very important. Use a clock or stopwatch.
- False readings will result if test strip is read too early or too late.

Test Intervals

Testing is recommended if operator is **not** sure of his cooling system condition due to leaks, uncontrolled topping off of system, or major coolant loss.

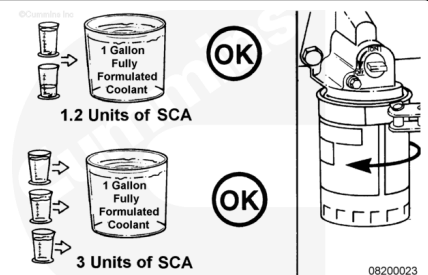
Testing is also recommended twice a year to monitor SCA level. If SCA level is above 3 units, test at subsequent oil drain intervals until concentration is back under 3 units. When concentration is back under 3 units, start installing correct service filters according to maintenance schedule used.



If concentration is below 1.2 units per 3.78 liter [1 gal], replace filter and precharge with liquid.

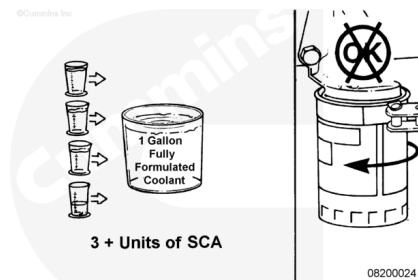


If concentration is 1.2 to 3 units per 3.78 liter [1 gal], replace filter.



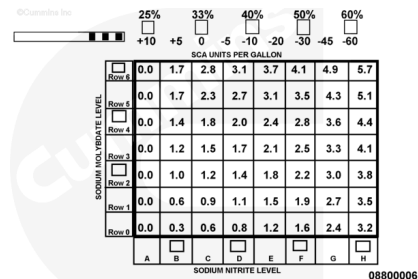
If concentration is above 3 units per 3.78 liter [1gal], do **not** replace service filter. Test coolant at subsequent oil drain intervals until concentration is back under 3 units. When

concentration is back under 3 units, start installing service filters at each oil change interval.



Note : Do not utilize the test kit to maintain minimum SCA concentration levels (i.e., 1.5 units).

Note : In some instances nitrate or molybdate reading will be high. However, it is combined reading that is important. Therefore, always follow chart.



Coolant Test Kit, Part Number CC2602

Works with any SCA formulation. Call 1-800-CUMMINS if you have this test kit and color chart does **not** show the number of units of SCA per gallon of coolant.

Probablizer:

Plug, Part Number 3318169S

Installs on engine for easy coolant sampling

Cap, Part Number 3318168S

Use with Monitor C™ bottle to sample coolant

Monitor C™, Part Number CC2700

Lab analysis of coolant samples

Contact Information

Cummins: 1-800-CUMMINS
1-800-521-4005

1-800-22-FILTERS
1-800-223-4583

Call above numbers to get answers to any questions you may have about cooling system maintenance.

Coolant Replacement Requirements

Drain and flush cooling system after 9000 hours or 2 years of service. Fill with either new fully formulated coolant or a 50/50 mixture of good quality water and fully formulated antifreeze, and install correct service coolant filter.

If coolant is **not** going to be reused, dispose of it in accordance with federal, state, and local laws and regulations.

